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The Influence of Proglacial Lakes on Ecological Succession in Proglacial Braidplains

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How does a glacier's retreat affect downstream environments?

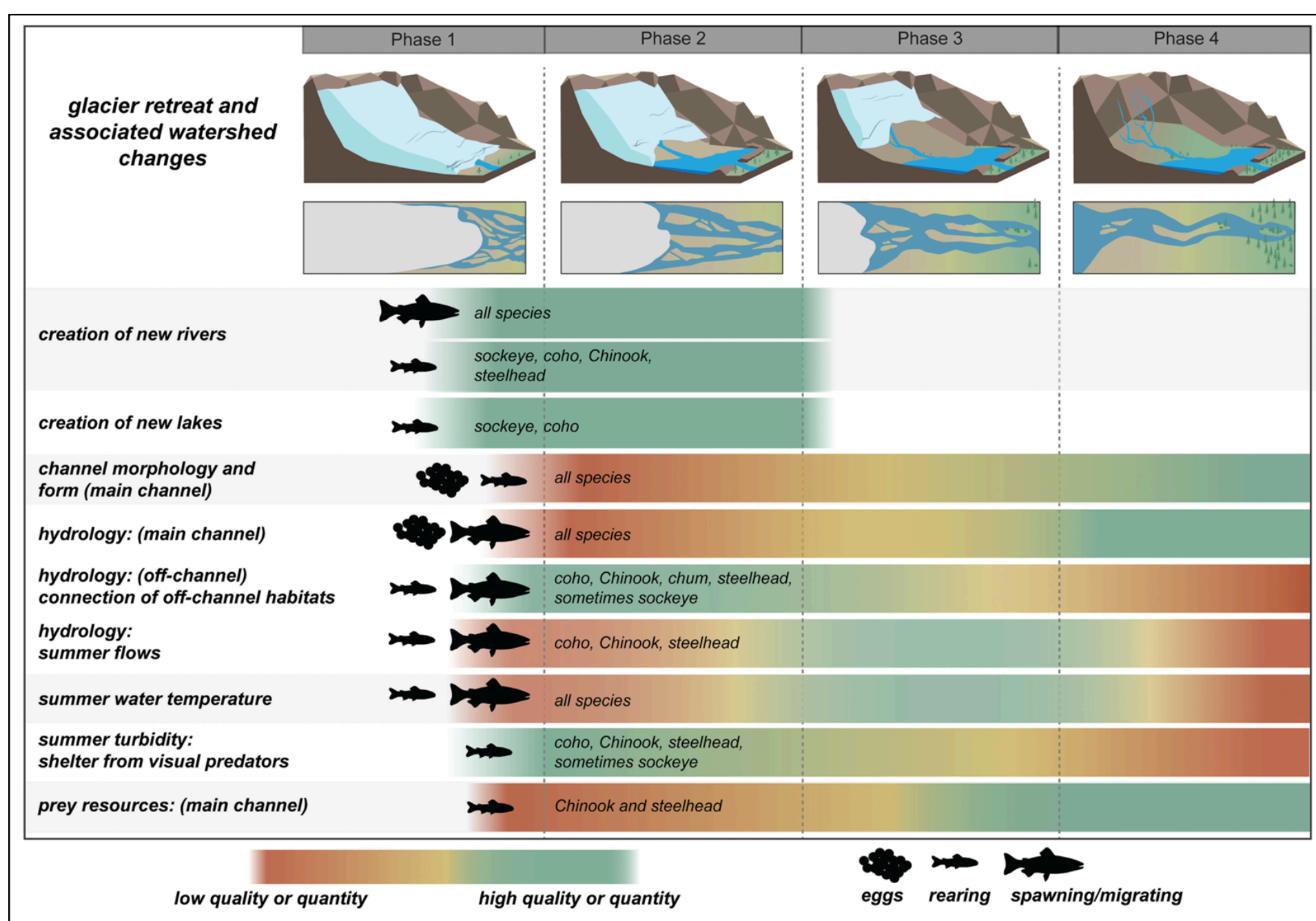


Figure 1: Retreating glacier's effect on downstream species in the Pacific Ocean (Pitman et al., 2020). Glacier retreat not only drives shifts in ecological succession within proglacial environments, but also alters the physical and thermal conditions of downstream waters. These changes can affect cold-water species such as salmon and other fish that depend on stable, low-temperature habitats fed by glacial meltwater.

Research Questions:
How is braidplain stability changing over time?

How does the formation of proglacial lakes alter downstream channel stability and ecological succession?

Hypothesis: Lake braidplains are expected to exhibit more rapid increases in NDVI through time, as greater channel stability promotes faster vegetation establishment compared to no-lake braidplains.

What does NDVI suggest about ecological succession?

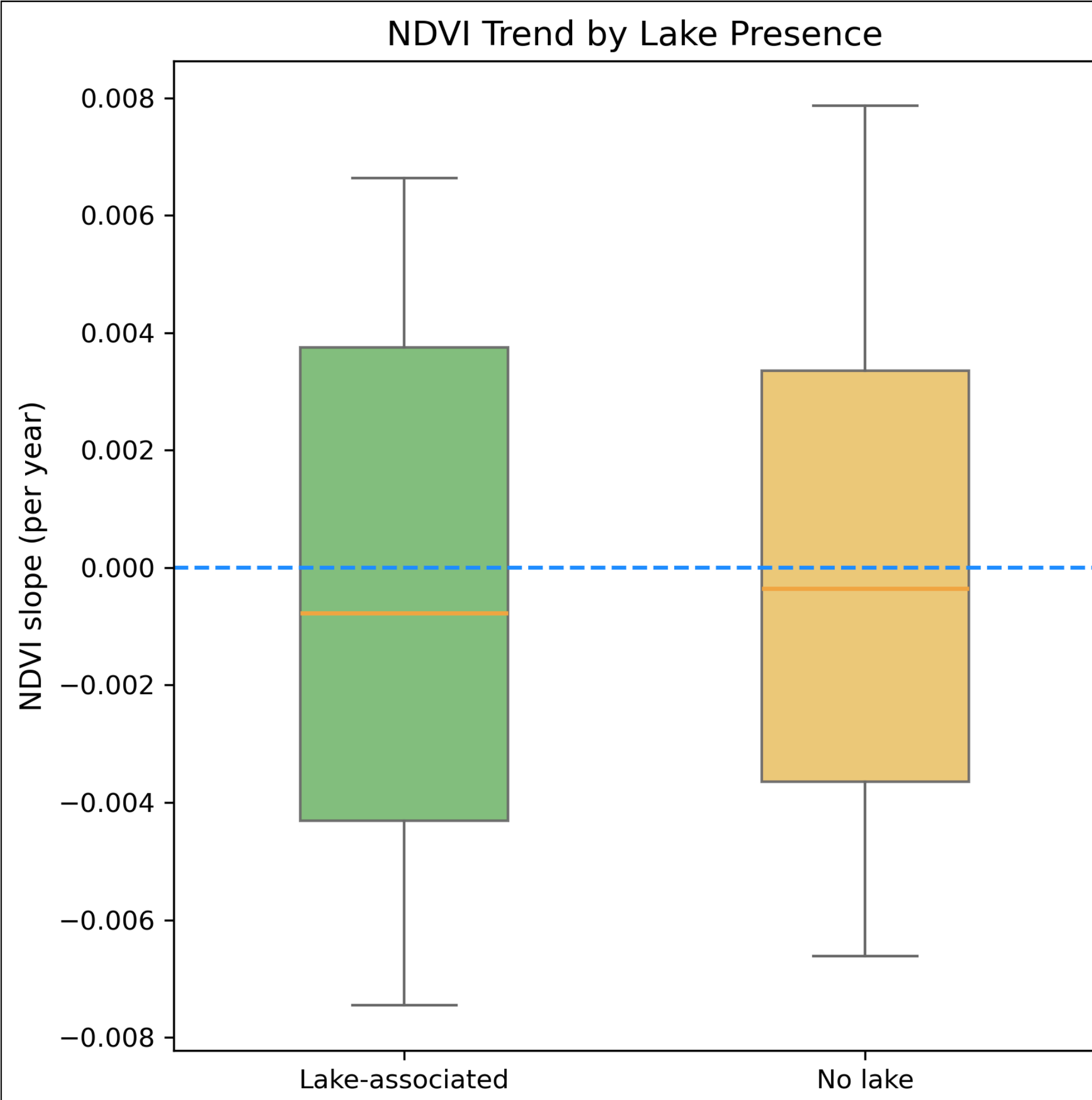


Figure 5: Boxplot shows the rate of NDVI slope per year for lake and no-lake braidplains.

Contrary to our hypothesis, stating that lake braidplains would exhibit faster increases in NDVI due to greater channel stability, no-lake braidplains display higher rates of NDVI increase.

Many proglacial lakes likely existed before the earliest Landsat images used in this study (1984), meaning they had already served as sediment traps for decades. If so, the downstream braidplains affected by these lakes would have experienced extended periods of moderated sediment flow, allowing for there to be established vegetation and extended ecological succession.

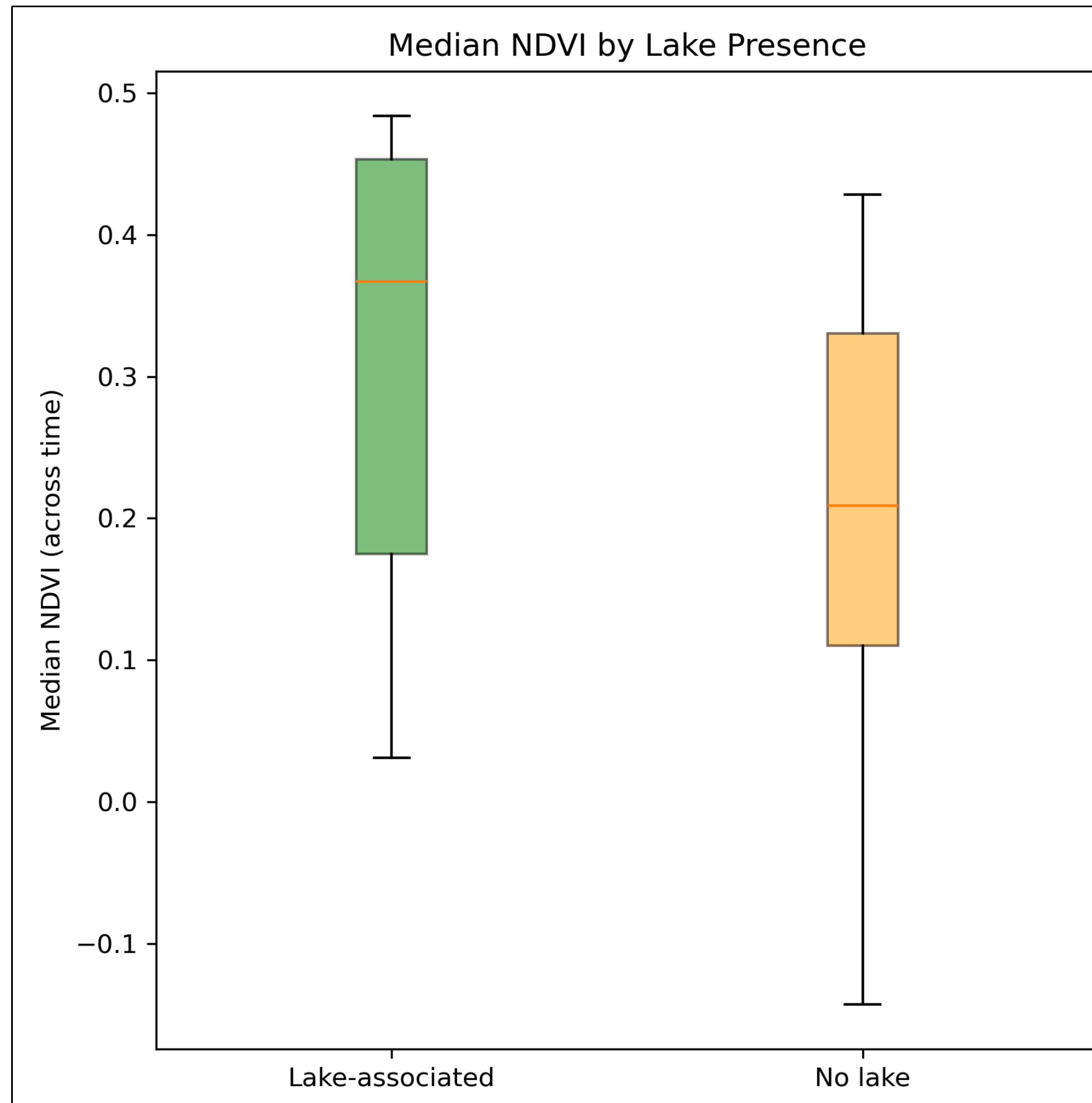


Figure 6: Distribution of median NDVI values across braidplains grouped by lake presence.

No-lake braidplains have a higher “greening” rate, but an overall lower NDVI, this may be caused by the saturation effect.

NDVI saturation can contribute to lower apparent NDVI change in lake braidplains.

In dense, well-established vegetation, NDVI becomes less sensitive to increases in biomass and canopy structure, causing values to plateau even as vegetation continues to develop (Gao et al., 2023).

Braidplains Across the Juneau Icefield

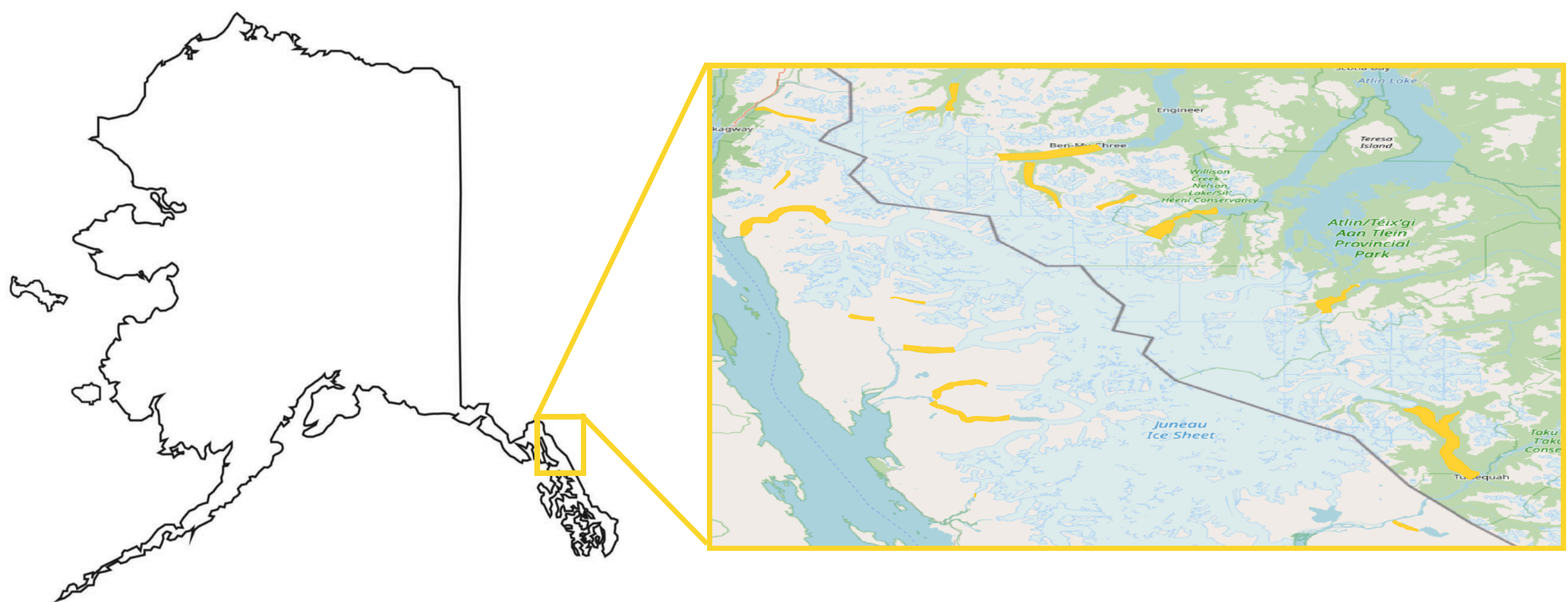


Figure 2: In Alaska, proglacial lakes increased by 38% in number and 59% in area between 1984 and 1988 and between 2016 and 2019, and glaciers that end in lakes account for 11% of global glacier area. Proglacial braidplains are highly active systems located between glaciers and downstream rivers. They feature high sediment loads and “braided” channels (Ashmore, 2013; Church & Ferguson, 2015). **These conditions create unstable landscapes where ongoing disturbances prevent the surface from stabilizing and hinder ecological succession** (Carrivick & Tweed, 2013).

Using Landsat and NDVI to Analyze Ecological Succession

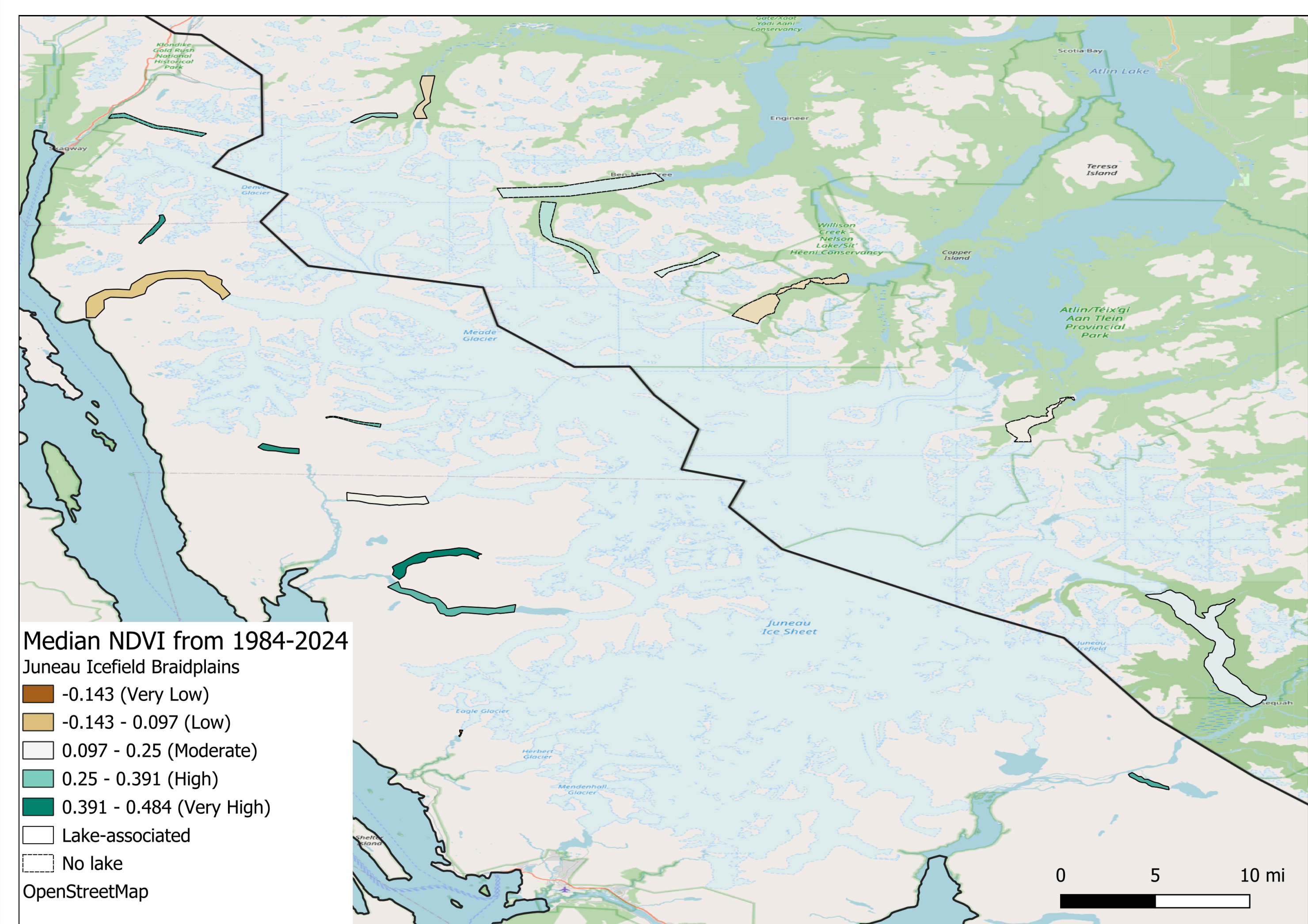


Figure 3: Median NDVI values for each braidplain were calculated from Landsat surface reflectance imagery obtained from the United States Geological Survey, spanning the 1980s through the 2020s.

Scenes were selected based on minimal cloud cover, consistent seasonal timing (July–August), and optimal spatial coverage of the study area across Landsat 5 TM and Landsat 8 OLI sensors. **The Normalized Difference Vegetation Index (NDVI), defined as $(NIR - Red) / (NIR + Red)$** , was computed for each time step and extracted within braidplain polygons using zonal statistics.

Median NDVI values represent long-term vegetation conditions within each braidplain, with **higher values indicating greater vegetation density**. Spatial patterns in median NDVI reflect variability in ecological development across the study area, where higher NDVI is interpreted as more established vegetation and increased landscape stability.

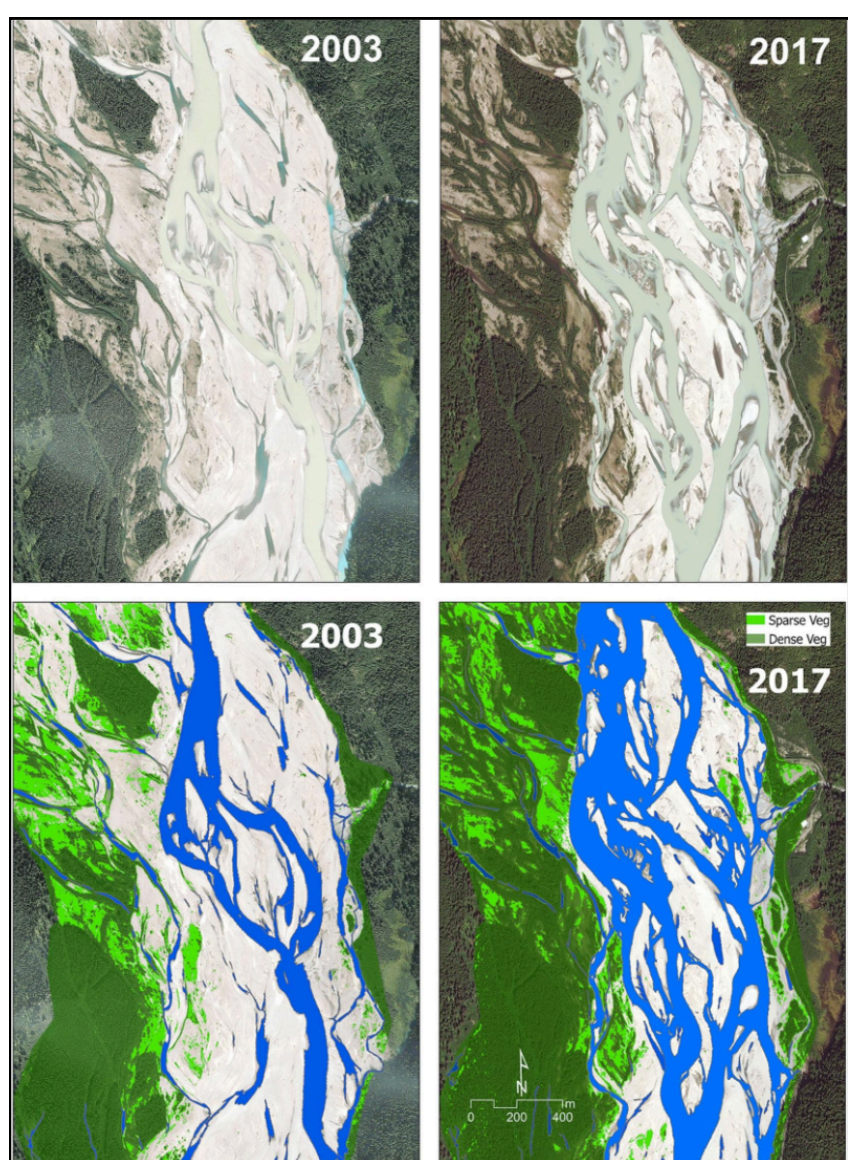


Figure 4: Comparison of sparse and dense vegetation using NDVI Analysis (Sergeant et al., 2023)

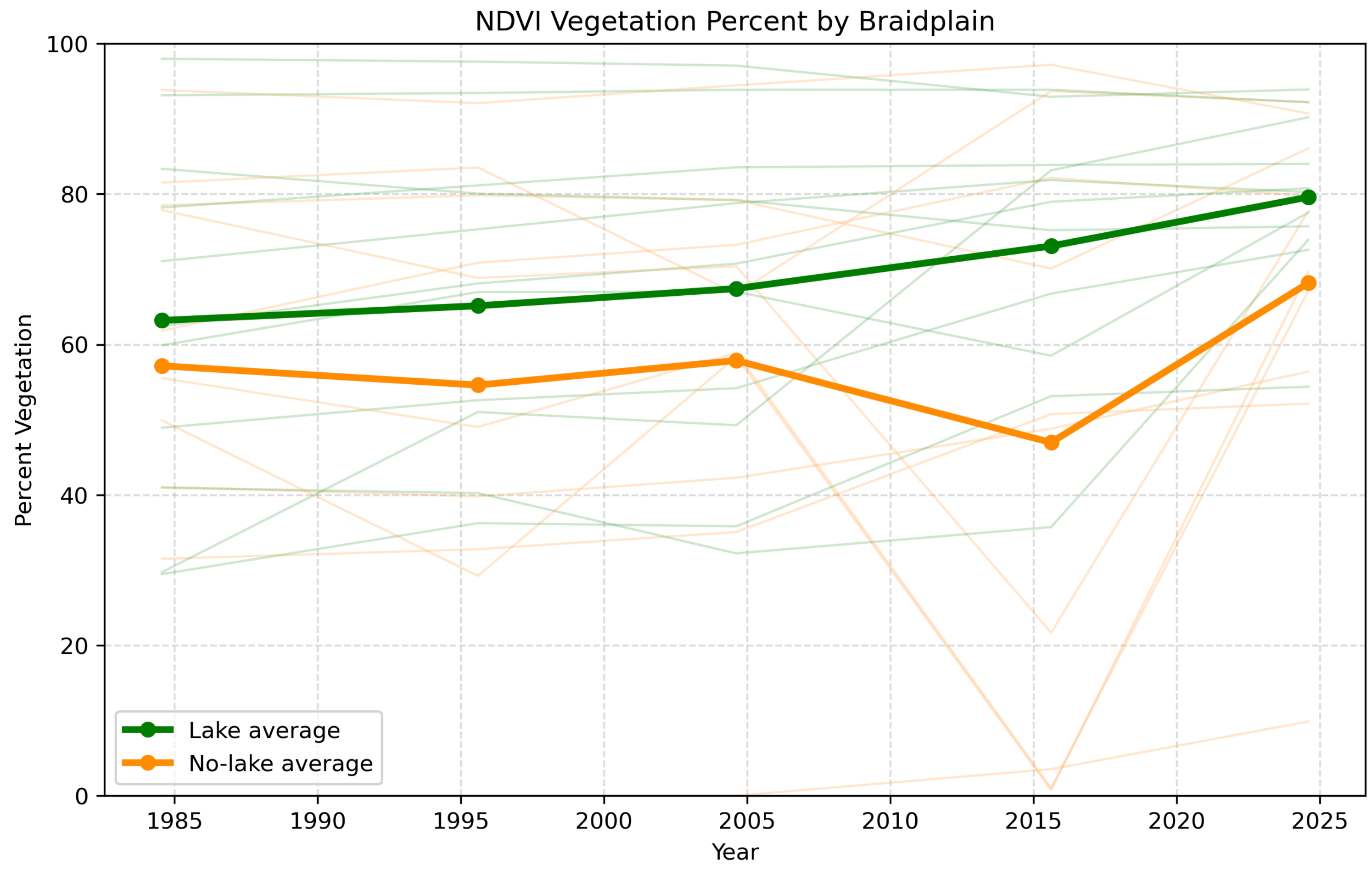


Figure 7: Percent vegetation across Juneau Icefield braidplains from 1984–2024, grouped by lake presence. Bold lines represent mean trends for lake (green) and no-lake (orange) braidplains, while lighter lines show individual braidplain trajectories. Both braidplain-types show overall increases in vegetation through time, indicating continued ecological succession across the study area.

Lake braidplains maintain higher vegetation cover, suggesting early stabilization prior to 1984. In contrast, no-lake braidplains show greater variability and stronger recent increases in vegetation, indicating ongoing stabilization and succession.

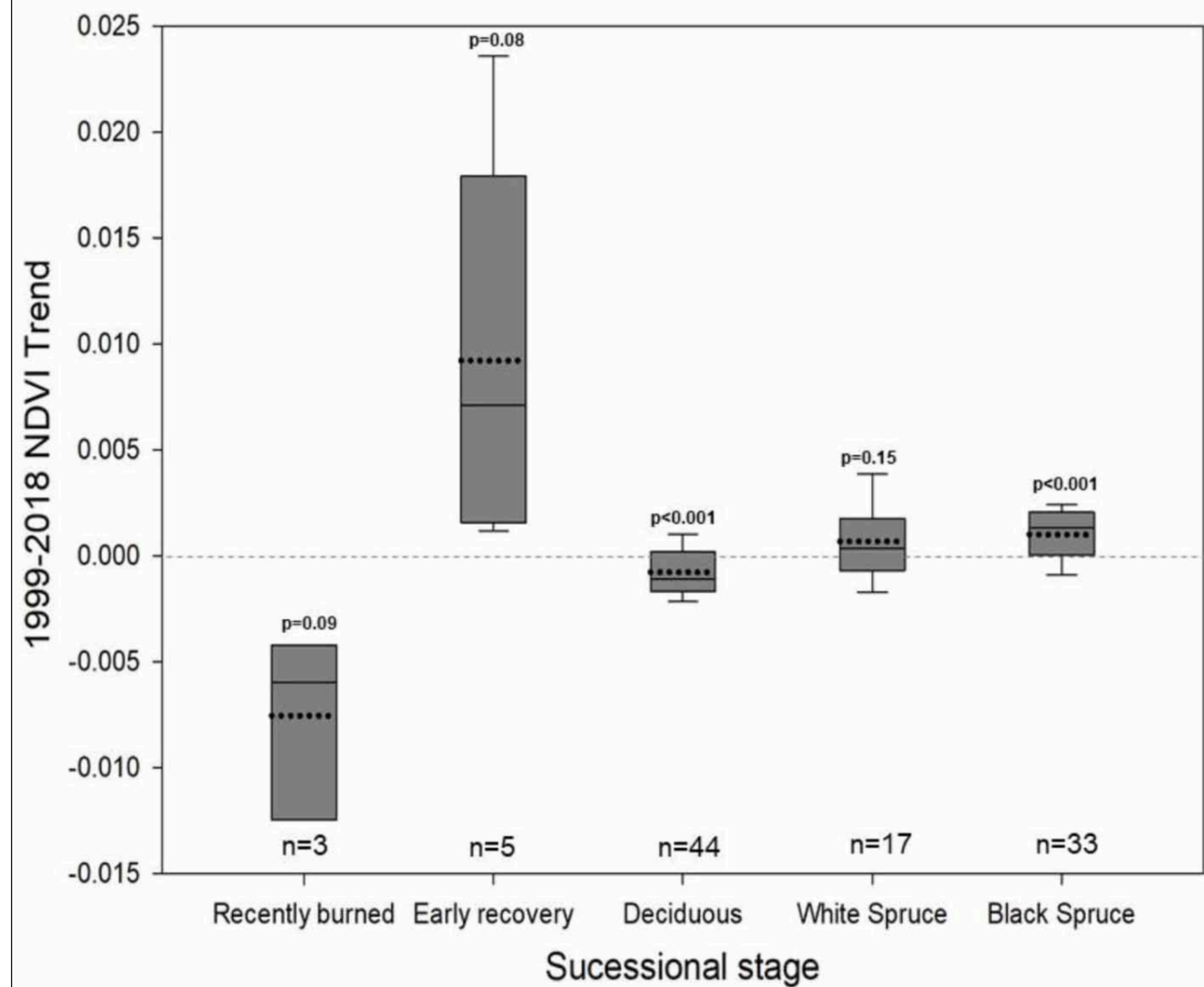


Figure 8: NDVI trends (1999–2018) across successional stages (Fiore et al., 2020), illustrating how vegetation greenness changes as ecosystems develop over time.

This pattern supports that **higher NDVI slopes in no-lake braidplains likely reflect active early succession**. Conversely, **lower NDVI trends observed in more stable, lake braidplains may indicate that these systems are already in later stages of ecological development**, with well-established vegetation and reduced rates of change through time.

Therefore, differences in NDVI trends across braidplains are better explained by successional stage and braidplain stability, rather than NDVI saturation effects alone.

Key Takeaways

- NDVI increases at a faster rate in no-lake braidplains, indicating more rapid vegetation establishment compared to lake braidplains.
- No-lake braidplains represent more actively evolving braidplains, leading to faster observable “greening,” while lake systems likely stabilized prior to the Landsat record.
- NDVI saturation in dense vegetation may limit detectable increases in vegetation growth in lake braidplains, masking continued ecological succession.
- Proglacial lakes do not impact the rate of vegetation change, but instead influence the timing of stabilization and ecological succession.

Future Work

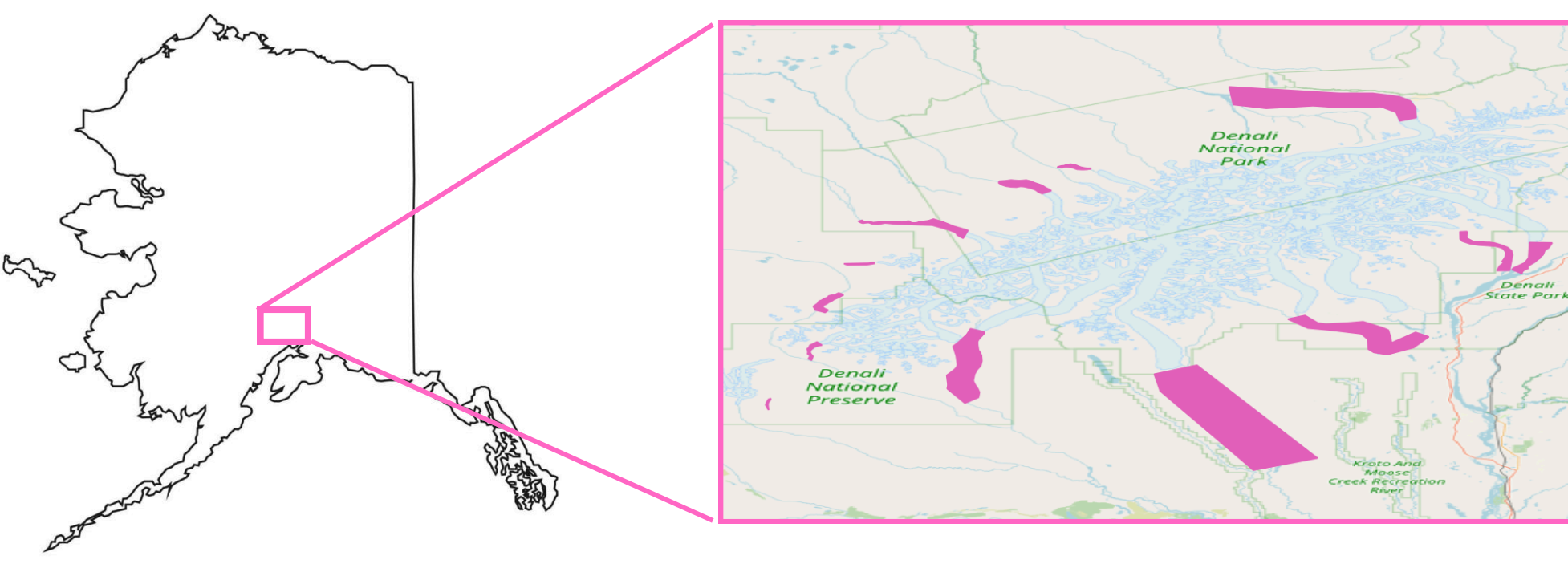


Figure 9: Future work will focus on Denali, which is characterized by continental, drier conditions compared to Juneau Icefield's maritime, wetter conditions.

Future research will focus on examining lake formation processes, assessing whether elevation impacts braidplain stability, and studying other climatic regions of Alaska to determine if specific climate regimes significantly influence braidplain stability. These findings will facilitate our understanding of the impact of climate change on the ecological succession within these regions, including whether stability will be maintained and if organisms will prosper to establish a vibrant and stable ecosystem capable of supporting Alaska's broader ecological network in a changing climate.

Acknowledgements and References

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Ashmore, P. (2013). 9.17 Morphology and dynamics of braided rivers. In Elsevier eBooks (pp. 289–312). <https://doi.org/10.1016/b978-0-12-374739-6.00242-6>
Carrivick, J. L., & Tweed, F. S. (2013a). Proglacial lakes: character, behaviour and geological importance. *Quaternary Science Reviews*, 78, 34–52. <https://doi.org/10.1016/j.quascirev.2013.07.028>
Church, M., & Ferguson, R. I. (2015). Morphodynamics: Rivers beyond steady state. *Water Resources Research*, 51(4), 1883–1897. <https://doi.org/10.1002/2014wr016862>
Fiore, N. M., Goulden, M. L., Czimczik, C. I., Pedron, S. A., & Tayo, M. A. (2020). Do recent NDVI trends demonstrate boreal forest decline in Alaska? *Environmental Research Letters*, 15(9), 095007. <https://doi.org/10.1088/1748-9326/ab9c4c>
Gao, S., Zhong, R., Yan, K., Ma, X., Chen, X., Pu, J., Gao, S., Qi, J., Yin, G., & Myneni, R. B. (2023). Evaluating the saturation effect of vegetation indices in forests using 3D radiative transfer simulations and satellite observations. *Remote Sensing of Environment*, 295, 113665. <https://doi.org/10.1016/j.rse.2023.113665>
Pitman, K. J., Moore, J. W., Sloat, M. R., Beaudreau, A. H., Bidlack, A. L., Brenner, R. E., Hood, E. W., Pess, G. R., Mantua, N. J., Milner, A. M., Radić, V., Reeves, G. H., Schindler, D. E., & Whited, D. C. (2020). Glacier retreat and Pacific salmon. *BioScience*, 70(3), 220–236. <https://doi.org/10.1093/biosci/biaa015>
Sergeant, C. J., Moore, J. W., Whited, D. C., Pitman, K. J., Connor, M., & Sexton, E. K. (2023). An interdisciplinary synthesis of floodplain ecosystem dynamics in a rapidly deglaciating watershed. *The Science of the Total Environment*, 912, 169245. <https://doi.org/10.1016/j.scitotenv.2023.169245>